

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with a strong foundation in statistical analysis (Python, R, Excel) and hands-on experience turning raw, imperfect data into clear findings for stakeholders. Enjoys finding patterns in data and communicating them in an accessible way, drawing on experience at Deutsche Bundesbank and in independent research and analytics projects.

Core Competencies

- **Statistical Analysis:** Python, R (certified, LinkedIn Learning), Excel; pattern detection and anomaly analysis in structured and unstructured data
- **Data Storytelling:** Structured reporting for both technical and non-technical stakeholders, translating findings from automated checks into clear, actionable summaries
- **Independent Analysis:** Self-directed data-collection and analysis projects, from healthcare risk profiling to air-quality classification
- **Visualization:** Matplotlib, Seaborn, Tableau; comfortable learning new analytics tools (e.g. Power BI) quickly
- **Databases & Tools:** SQL, SQLite, MySQL, Excel, Git
- **Languages:** German (C1), English (C1, TOEFL iBT 105/120), Spanish (native)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations, with automated Jira ticketing and structured reporting to reporting banks
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Projects

- **Tecnológico de Monterrey** 10/2024–12/2024
Predictive Modeling – Healthcare Optimization
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks; built predictive models and K-Means clustering for high-risk segmentation; conducted time-series and regional analyses to uncover seasonal patterns
- **Tecnológico de Monterrey** 04/2023–06/2023
Mexico City Air Quality Analysis – SVM Classification
 - Analyzed city-wide NO_x/O₃ data and applied SVM models to classify critical ozone levels; correlation analysis and visualizations identified elevated pollution patterns by district; findings presented in a final presentation
- **Personal Project (GitHub)** 05/2026
Banking Data Pipeline – Data Cleaning & Audit Log
 - Built an end-to-end pipeline for simulated bank transactions with a normalized SQLite database, staged data cleaning, and a structured exception report

Education

- **Tecnológico de Monterrey** **Mexico**
2022–2027
BSc in Data Science
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, numerical optimization.
- **Universität Göttingen** **Germany**
2025–2026
Exchange Semester

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Certifications

- R Programming – LinkedIn Learning (completed 05/2026).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).

References

- Available upon request.